

Ma trận với định thức

Bài 1.1

$$A = \begin{pmatrix} 2 & -1 \\ 5 & -2 \end{pmatrix}$$

$$a, A^{567} = ?$$

$$\text{Ta có } A^2 = \begin{pmatrix} 2 & -1 \\ 5 & -2 \end{pmatrix} \begin{pmatrix} 2 & -1 \\ 5 & -2 \end{pmatrix} = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix} = -I$$

$$A^{567} = A^{566} \cdot A = (A^2)^{283} \cdot A = -A$$

b

$$\det(A^{576} + 2A^{567} + 3A^{675}) = ?$$

Ta có

$$A^{576} + 2A^{567} + 3A^{675}$$

$$A^{576} = (A^2)^{288} = I$$

$$A^{567} = -A \rightarrow 2A^{567} = -2A$$

$$3A^{675} = 3(A^2)^{337} \cdot A = -3A$$

$$\rightarrow A^{576} + 2A^{567} + 3A^{675} = I - 5A$$

$$= \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} - \begin{pmatrix} 10 & -5 \\ 25 & -10 \end{pmatrix}$$

$$= \begin{pmatrix} -9 & 5 \\ -25 & 11 \end{pmatrix}$$

$$\rightarrow \det(A^{576} + 2A^{567} + 3A^{675}) = \begin{vmatrix} -9 & 5 \\ -25 & 11 \end{vmatrix} = 26$$

Bài 1.2:

$$a, \text{ Ta có } A^2 = A \cdot A = \begin{pmatrix} 2 & 3 \\ -1 & -1 \end{pmatrix} \cdot \begin{pmatrix} 2 & 3 \\ -1 & -1 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 3 \\ -1 & -2 \end{pmatrix}$$

$$A^{2018} = A^2 = \begin{pmatrix} 1 & 3 \\ -1 & -2 \end{pmatrix}$$

$$b, \det(2A^{2017} - 3A^{2018} + 4A^{2019})$$

Ta có

$$2A^{2017} = 2A^{2016} \cdot A = 2(A^2)^{1008} \cdot A = 2A^2 \cdot A = 2A^3$$

$$4A^{2019} = 4A^{2018} \cdot A = 4A^3$$

$$\rightarrow 2A^{2017} - 3A^{2018} + 4A^{2019} = 2A^3 - 3A^2 + 4A^3$$

$$= 6A^3 - 3A^2$$

$$= \begin{pmatrix} -6 & 0 \\ 0 & -6 \end{pmatrix} - \begin{pmatrix} 3 & 9 \\ 3 & -6 \end{pmatrix}$$

$$= \begin{pmatrix} -9 & -9 \\ 3 & 0 \end{pmatrix}$$

$$\det(\dots) = \begin{vmatrix} -9 & -9 \\ 3 & 0 \end{vmatrix} = 27$$

Bài 1.3

$$A = \begin{pmatrix} 1 & -4 & 2 \\ 1 & -4 & 2 \\ 1 & -4 & 2 \end{pmatrix}$$

$$\text{Ta có } A^2 = A \cdot A$$

$$= \begin{pmatrix} -1 & 4 & -2 \\ -1 & 4 & -2 \\ -1 & 4 & -2 \end{pmatrix} = -A$$

$$A^{200} + A = A^2(-1)^{100} + A = \begin{pmatrix} -1 & 4 & -2 \\ -1 & 4 & -2 \\ -1 & 4 & -2 \end{pmatrix} + \begin{pmatrix} 1 & -4 & 2 \\ 1 & -4 & 2 \\ 1 & -4 & 2 \end{pmatrix}$$

$$= \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} = \mathbf{0}$$

Bài 1.4

$$A = \begin{pmatrix} -10 & 11 & -22 \\ 3 & -2 & 6 \\ 6 & -6 & 13 \end{pmatrix}$$

a) Tính A^2 , A^{2018} , A^{2019}

$$A^2 = A \cdot A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = I$$

$$A^{2018} = (A^2)^{1009} = I$$

$$A^{2019} = A^{2018} \cdot A = I \cdot A = A$$

b)

Bài 1.5

$$A = \begin{pmatrix} 1 & 0 & a \\ 0 & 1 & b \\ 0 & 0 & -1 \end{pmatrix}$$

a) A^2 , A^{2018} , A^{2019}

$$A^2 = A \cdot A = \begin{pmatrix} 1 & 0 & a \\ 0 & 1 & b \\ 0 & 0 & -1 \end{pmatrix} \begin{pmatrix} 1 & 0 & a \\ 0 & 1 & b \\ 0 & 0 & -1 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = I$$

$$A^{2018} = (A^2)^{1009} = I$$

$$A^{2019} = A^{2018} A = I \cdot A = A$$

Dải 1.6

$$\begin{vmatrix} 3 & 3 & x & x \\ x & 3 & 3 & x \\ x & x & 3 & 3 \\ x & x & x & 3 \end{vmatrix} = 0$$

$$\text{Ta có } \begin{vmatrix} 3 & 3 & x & x \\ x & 3 & 3 & x \\ x & x & 3 & 3 \\ x & x & x & 3 \end{vmatrix} = 3 \cdot A_{12} + 3 \cdot A_{22} + x \cdot A_{32} + x \cdot A_{42}$$

$$= 3 \cdot (-1)^3 \cdot \begin{vmatrix} x & 3 & x \\ x & 3 & 3 \\ x & x & 3 \end{vmatrix} + 3 \cdot \begin{vmatrix} 3 & x & x \\ x & 3 & 3 \\ x & x & 3 \end{vmatrix} + x \cdot (-1)^5 \cdot \begin{vmatrix} 3 & x & x \\ x & 3 & 3 \\ x & x & 3 \end{vmatrix}$$

$$+ x \cdot \begin{vmatrix} 3 & x & x \\ x & 3 & x \\ x & x & 3 \end{vmatrix}$$

$$= -3(9x + x^3 + 9x - (3x^2 + 9x + 3x^2)) + 3(27 + x^3 + 3x^2 - (3x^2 + 3x^2 + 3x^2))$$

$$- x[27 + x^3 + x^3 - (3x^2 + 3x^2 + 3x^2)] + x[27 + 3x^2 + x^3 - (3x^2 + 3x^2 + 9x)]$$

$$= -3(x^3 - 6x^2 + 9x) + 3(x^3 + 3x^2 + 9x + 27)$$

$$- x(2x^3 - 9x^2 + 27) + x(x^3 - 3x^2 + 9x + 27)$$

$$= -2x^4 + 9x^3 - 27x + x^4 - 3x^3 + 9x^2 + 27x + 81$$

HAI TIẾN

$$-x^4 + 8x^3 + \cancel{8x^2} + 81 = 0$$

$$\Leftrightarrow \begin{cases} x = 3 \\ x = -3 \end{cases}$$

1.11

$$A = \begin{pmatrix} 4 & 2 \\ 1 & 3 \end{pmatrix}$$

$$B = \begin{pmatrix} 3 & 1 \\ 2 & 3 \end{pmatrix}$$

$$\begin{aligned} a) & \det(A^3 B^2 + 4A^2 B^3) \\ &= \det[A^2(A \cdot B^2 + 4B^3)] \\ &= \det[A^2(A + 4B)B^2] \end{aligned}$$

$$\begin{aligned} &= (\det A)^2 \cdot \det(A + 4B) \cdot (\det B)^2 \\ &= 100 \cdot \begin{vmatrix} 10 & 6 \\ 9 & 15 \end{vmatrix} \cdot 49 = 91400 \end{aligned}$$

$$\begin{aligned} b) \text{ Tính } A + 2B &= \begin{pmatrix} 4 & 2 \\ 1 & 3 \end{pmatrix} + 2 \begin{pmatrix} 3 & 1 \\ 2 & 3 \end{pmatrix} \\ &= \begin{pmatrix} 10 & 4 \\ 5 & 9 \end{pmatrix} \end{aligned}$$

$$\text{Ta có } (A + 2B)^2 - 19(A + 2B) = \begin{pmatrix} -70 & 0 \\ 0 & -70 \end{pmatrix} = -70I$$

$$\begin{pmatrix} 4 & 3 \\ 3 & 2 \end{pmatrix} \times \begin{pmatrix} 7 & 5 \\ 3 & 2 \end{pmatrix} = \begin{pmatrix} -1 & 2 \\ -1 & 0 \end{pmatrix}$$

$$12) \text{ Với } A = \begin{pmatrix} 4 & 3 \\ 3 & 2 \end{pmatrix} \quad B = \begin{pmatrix} 7 & 5 \\ 3 & 2 \end{pmatrix} \quad C = \begin{pmatrix} 1 & 2 \\ -1 & 0 \end{pmatrix}$$

$$A^{-1} = \frac{1}{\det A} A^* = \frac{1}{-1} \begin{pmatrix} 2 & -3 \\ -3 & 4 \end{pmatrix} = \begin{pmatrix} -2 & 3 \\ 3 & -4 \end{pmatrix}$$

$$B^{-1} = \frac{1}{\det B} B^* = \frac{1}{-1} \begin{pmatrix} 2 & -5 \\ -3 & 7 \end{pmatrix} = \begin{pmatrix} -2 & 5 \\ 3 & -7 \end{pmatrix}$$

HÀI TIẾN

$$\Rightarrow \text{pt} \Leftrightarrow A \cdot X \cdot B = C \Leftrightarrow A^{-1} \cdot A \cdot X \cdot B \cdot B^{-1} = A^{-1} C B^{-1}$$

$$X = A^{-1} C B^{-1}$$

$$= \begin{pmatrix} -2 & +3 \\ +3 & -4 \end{pmatrix} \begin{pmatrix} 1 & 2 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} -2 & 5 \\ 3 & -7 \end{pmatrix}$$

$$= \begin{pmatrix} -2 & 3 \\ 4 & -7 \end{pmatrix}$$

1.25

$$A = \begin{pmatrix} 1 & 1 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 6 & 4 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} 1 & 1 & 2 & 1 \\ 0 & -1 & 0 & 1 \\ 0 & 3 & 2 & 6 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} 1 & 1 & 2 & 1 \\ 0 & -1 & 0 & 1 \\ 0 & -1 & 0 & 1 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 1 & 2 & 1 \\ 0 & -1 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix} = B$$

$$\text{Ta có } r(A) = r(B) = 2$$

1.26

$$A = \begin{pmatrix} 1 & -1 & 3 & 2 & -1 \\ 2 & 2 & -1 & 2 & 3 \\ 1 & -2 & -2 & 2 & -4 \\ 4 & -1 & 0 & 6 & -2 \end{pmatrix} \begin{array}{l} H_1(-2) + H_2 \rightarrow H_2 \\ H_4(-1) + H_3 \rightarrow H_3 \\ H_4(-4) + H_5 \rightarrow H_5 \end{array}$$

$$\Rightarrow \begin{pmatrix} 1 & -1 & 3 & 2 & -1 \\ 0 & 4 & -7 & -2 & 5 \\ 0 & -1 & -5 & 0 & -3 \\ 0 & -3 & -12 & -2 & 2 \end{pmatrix} \begin{array}{l} H_2 + H_3(+4) \rightarrow H_2 \\ H_2(-3) + H_4(4) \rightarrow H_4 \end{array}$$

$$\Rightarrow \begin{pmatrix} 1 & -1 & 3 & 2 & -1 \\ 0 & 4 & -7 & -2 & 5 \\ 0 & 0 & -27 & -2 & -7 \\ 0 & 0 & -27 & -2 & -7 \end{pmatrix} \begin{array}{l} H_3(-1) + H_4 \rightarrow H_4 \end{array}$$

$$\rightarrow \begin{pmatrix} 1 & -1 & 3 & 2 & -1 \\ 0 & 4 & -7 & -2 & 5 \\ 0 & 0 & -27 & -2 & -7 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix} = B$$

$$\text{Ta có } \lambda(A) = \lambda(B) = 3$$

1.19.

$$A = \begin{pmatrix} x & 2 & 3 \\ 0 & 1 & 4 \\ 0 & 5 & 8 \end{pmatrix}$$

$$\text{Đề' } A^4 - 3A^3 \text{ khả nghịch} \Leftrightarrow \det(A^4 - 3A^3) \neq 0 \\ \Leftrightarrow \det[A^3(A - 3I)] \neq 0$$

$$\text{Ta có } \det A = x \begin{vmatrix} -1 & 1 \\ 5 & 8 \end{vmatrix} = x \begin{vmatrix} 1 & 4 \\ 5 & 8 \end{vmatrix} \\ = -12x$$

$$\Rightarrow A - 3I = \begin{pmatrix} x & 2 & 3 \\ 0 & 1 & 4 \\ 0 & 5 & 8 \end{pmatrix} - 3 \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \\ = \begin{pmatrix} x-3 & 2 & 3 \\ 0 & 2 & 4 \\ 0 & 5 & 5 \end{pmatrix}$$

$$\det(A - 3I) = \begin{vmatrix} x-3 & 2 & 3 \\ 0 & 2 & 4 \\ 0 & 5 & 5 \end{vmatrix} = (x-3) \begin{vmatrix} -1 & 1 \\ 5 & 5 \end{vmatrix} \\ = -30(x-3)$$

$$\Rightarrow (\det A)^3 \cdot \det(A - 3I) \neq 0$$

$$\Leftrightarrow \begin{cases} \det A \neq 0 \\ \det(A - 3I) \neq 0 \end{cases} \Leftrightarrow \begin{cases} -12x \neq 0 \\ -30(x-3) \neq 0 \end{cases} \Leftrightarrow \begin{cases} x \neq 0 \\ x \neq 3 \end{cases}$$

1.30

$$A = \begin{pmatrix} 1 & x & x & x \\ 1 & 1 & x & x \\ 1 & x & 2 & x \\ 1 & 1 & 2 & 2 \end{pmatrix}$$

$$\longrightarrow \begin{pmatrix} 1 & x & x & x \\ 0 & x-1 & 0 & 0 \\ 0 & 0 & x-2 & 0 \\ 0 & x-1 & x-2 & x-2 \end{pmatrix}$$

$$\longrightarrow \begin{pmatrix} 1 & x & x & x \\ 0 & x-1 & 0 & 0 \\ 0 & 0 & x-2 & 0 \\ 0 & 0 & 2-x & 2-x \end{pmatrix}$$

$$\text{Để } \lambda(A) = 2 \Leftrightarrow x-2 = 0 \Leftrightarrow x = 2$$

1.29

$$A = \begin{pmatrix} 2 & x & x & x \\ x & 2 & x & x \\ x & x & 2 & x \end{pmatrix} \longrightarrow \begin{pmatrix} x & x & x & 2 \\ x & 2 & x & x \\ x & x & 2 & x \end{pmatrix}$$

$$\longrightarrow \begin{pmatrix} x & x & x & 2 \\ -0 & x-2 & 0 & 2-x \\ 0 & 0 & x-2 & 2-x \end{pmatrix}$$

Ta có:

Nếu $x = 2$ thì $\lambda(A) = 1$ Nếu $x \neq 2$ thì $\lambda(A) = 3$

1.12.

$$A = \begin{pmatrix} 1 & 2 & 4 \\ 2 & 1 & -2 \\ 3 & 2 & 1 \end{pmatrix} \quad B = \begin{pmatrix} 3 & 4 & 5 \\ 2 & 2 & 3 \\ 4 & -1 & 3 \end{pmatrix}$$

$$\det(A \cdot B) = \det A \cdot \det B$$

$$= \begin{vmatrix} 1 & 2 & 4 \\ 2 & 1 & -2 \\ 3 & 2 & 1 \end{vmatrix} \cdot \begin{vmatrix} 3 & 4 & 5 \\ 2 & 2 & 3 \\ 4 & -1 & 3 \end{vmatrix}$$

$$= -47 \cdot 1 = -47$$

1.16

$$A = \begin{pmatrix} 2 & 1 & -3 \\ 1 & 3 & -2 \\ -1 & -2 & 1 \end{pmatrix}$$

$$\text{ta có } \det A = -4 \neq 0 \rightarrow \exists A^{-1}$$

$$A_{11} = (-1)^2 \begin{vmatrix} 3 & -2 \\ -2 & 1 \end{vmatrix} = 7$$

$$A_{32} = (-1)^5 \begin{vmatrix} 2 & -3 \\ 1 & -2 \end{vmatrix} = 1$$

$$A_{12} = (-1)^3 \begin{vmatrix} 1 & -2 \\ -1 & 1 \end{vmatrix} = +1$$

$$A_{33} = (-1)^6 \begin{vmatrix} 2 & 1 \\ 1 & 3 \end{vmatrix} = 5$$

$$A_{13} = (-1)^4 \begin{vmatrix} 1 & 3 \\ -1 & -2 \end{vmatrix} = 1$$

$$A_{21} = (-1)^3 \begin{vmatrix} 1 & -3 \\ -2 & 1 \end{vmatrix} = 5$$

$$A_{22} = (-1)^4 \begin{vmatrix} 2 & -3 \\ -1 & 1 \end{vmatrix} = -1$$

$$A_{23} = (-1)^5 \begin{vmatrix} 2 & 1 \\ -1 & -2 \end{vmatrix} = 3$$

$$A_{31} = (-1)^4 \begin{vmatrix} 1 & -3 \\ 3 & 2 \end{vmatrix} = 7$$

HÀ TIẾN

$$\text{Vậy } A^{-1} = \frac{1}{\det A} \cdot A^* \\ = \frac{1}{-4} \cdot \begin{pmatrix} 7 & 5 & 7 \\ 1 & -1 & 1 \\ 1 & 3 & 5 \end{pmatrix}$$

1.17,

$$A = \begin{pmatrix} 2 & 0 & 3 \\ 1 & 2 & 2 \\ 1 & 0 & 4 \end{pmatrix}$$

$$\text{a) Vậy } A^2 = A \cdot A = \begin{pmatrix} 2 & 0 & 3 \\ 1 & 2 & 2 \\ 1 & 0 & 4 \end{pmatrix} \cdot \begin{pmatrix} 2 & 0 & 3 \\ 1 & 2 & 2 \\ 1 & 0 & 4 \end{pmatrix} \\ = \begin{pmatrix} 7 & 0 & 18 \\ 6 & 4 & 15 \\ 6 & 0 & 19 \end{pmatrix}$$

$$A^3 = A^2 \cdot A = \begin{pmatrix} 7 & 0 & 18 \\ 6 & 4 & 15 \\ 6 & 0 & 19 \end{pmatrix} \cdot \begin{pmatrix} 2 & 0 & 3 \\ 1 & 2 & 2 \\ 1 & 0 & 4 \end{pmatrix} \\ = \begin{pmatrix} 32 & 0 & 93 \\ 31 & 8 & 86 \\ 31 & 0 & 94 \end{pmatrix}$$

$$A^3 - 8A^2 + 17A = \begin{pmatrix} 32 & 0 & 93 \\ 31 & 8 & 86 \\ 31 & 0 & 94 \end{pmatrix} - 8 \begin{pmatrix} 7 & 0 & 18 \\ 6 & 4 & 15 \\ 6 & 0 & 19 \end{pmatrix} + 17 \begin{pmatrix} 2 & 0 & 3 \\ 1 & 2 & 2 \\ 1 & 0 & 4 \end{pmatrix} \\ = \begin{pmatrix} 10 & 0 & 0 \\ 0 & 10 & 0 \\ 0 & 0 & 10 \end{pmatrix} = 10I$$

$$\text{b) Vậy } \det A = 10 \neq 0 \rightarrow \exists A^{-1}$$

$$A_{11} = (-1)^2 \cdot \begin{vmatrix} 2 & 2 \\ 0 & 4 \end{vmatrix} = 8 \quad A_{12} = (-1)^3 \cdot \begin{vmatrix} 1 & 2 \\ 1 & 4 \end{vmatrix} = -2$$

$$A_{13} = (-1)^4 \begin{vmatrix} 1 & 2 \\ 1 & 0 \end{vmatrix} = -2$$

$$A_{21} = (-1)^3 \begin{vmatrix} 0 & 3 \\ 0 & 4 \end{vmatrix} = 0$$

$$A_{22} = (-1)^4 \begin{vmatrix} 2 & 3 \\ 1 & 4 \end{vmatrix} = 5$$

$$A_{23} = (-1)^5 \begin{vmatrix} 2 & 0 \\ 1 & 0 \end{vmatrix} = 0$$

$$A_{31} = (-1)^4 \begin{vmatrix} 0 & 3 \\ 2 & 2 \end{vmatrix} = -6$$

$$A_{32} = (-1)^5 \begin{vmatrix} 2 & 3 \\ 1 & 2 \end{vmatrix} = -1$$

$$A_{33} = (-1)^6 \begin{vmatrix} 2 & 0 \\ 1 & 2 \end{vmatrix} = 4$$

Ta có $\det A = 10$

$$A^{-1} = \frac{1}{\det A} \cdot A^* = \frac{1}{10} \begin{pmatrix} 8 & 0 & -6 \\ -2 & 5 & -1 \\ -2 & 0 & 4 \end{pmatrix}$$

1.92

$$\begin{pmatrix} 1 & 0 & -1 & 4 \\ 2 & 1 & -1 \\ 1 & -2 & 1 \end{pmatrix} \quad X = \begin{pmatrix} 5 & 1 & 3 \\ 2 & 2 & -2 \\ 4 & -2 & 1 \end{pmatrix}$$

$A \qquad B$

$$\det A = -8 \neq 0 \Rightarrow \exists A^{-1}$$

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$$A \cdot X = B$$

$$\Leftrightarrow X = A^{-1} \cdot B$$

HÀ TIẾN

$$A_{11} = -1$$

$$A_{21} = -7$$

$$A_{31} = -3$$

$$A_{12} = -3$$

$$A_{22} = -3$$

$$A_{32} = 9$$

$$A_{13} = -5$$

$$A_{23} = 1$$

$$A_{33} = 3$$

$$A^{-1} = \frac{1}{\det A} \cdot A^* = \frac{1}{-18} \cdot \begin{pmatrix} -1 & -7 & -3 \\ -3 & -3 & 9 \\ -5 & 1 & 3 \end{pmatrix}$$

$$X = -A^{-1} \cdot B$$

$$= \frac{1}{-18} \begin{pmatrix} -1 & -7 & -3 \\ -3 & -3 & 9 \\ -5 & 1 & 3 \end{pmatrix} \cdot \begin{pmatrix} 5 & 1 & 3 \\ 2 & 2 & -2 \\ 4 & -2 & 1 \end{pmatrix}$$

$$= -\frac{1}{18} \begin{pmatrix} -31 & -9 & 8 \\ 15 & -27 & 6 \\ -11 & -9 & -14 \end{pmatrix}$$

1. 14 >

$$A = \begin{pmatrix} 1 & 2 & 5 \\ 3 & 4 & -1 \\ 4 & 2 & -3 \end{pmatrix}$$

$$B = \begin{pmatrix} 3 & 2 & -2 \\ 3 & 1 & 4 \\ 5 & 2 & 7 \end{pmatrix}$$

$$\begin{aligned} a, \det(A^3 B^2 - 3A^2 B^3) \\ &= \det A^2 (AB^2 - 3B^3) \\ &= (\det A^2) \cdot \det(A - 3B) \cdot (\det B^2) \end{aligned}$$

$$\det A = -50$$

$$\det B = -7$$

$$\det(A - 3B) = 997$$

$$\text{Vậy } \det(A^3 B^2 - 3A^2 B^3) = 122132500$$

1.13

$$A = \begin{pmatrix} 3 & 5 & 7 \\ 2 & 3 & -2 \\ 2 & -2 & 3 \end{pmatrix}$$

$$B = \begin{pmatrix} 1 & 4 & -5 \\ -2 & 2 & 3 \\ 4 & -1 & 2 \end{pmatrix}$$

$$\det(A^2B - 3AB^2)$$

$$= \det A (AB - 3AB^2)$$

$$= \det A \cdot \det(A - 3B) \det B$$

$$= -105 \cdot -1422 \cdot 101$$

$$= 15080310$$

Câu 1

$$A = \begin{pmatrix} 1 & 3 \\ 2 & 1 \end{pmatrix}$$

$$x^2 + 2x |A| + |A^2| = 0$$

$$\text{Ta có } |A| = 1 - 6 = -5$$

$$A^2 = \begin{pmatrix} 7 & 6 \\ 4 & 7 \end{pmatrix}$$

$$|A^2| = 25$$

$$\text{Pt} \Rightarrow x^2 - 10x + 25 = 0$$

$$\Rightarrow x = 5$$

Câu 2

$$\begin{vmatrix} x & 1 & -1 \\ 0 & x & 2 \\ 0 & 2 & x \end{vmatrix} = 0$$

$$\Leftrightarrow x \cdot (-1)^{1+1} A_{11} + 0 \cdot (-1) A_{12} + (-1) \cdot 0 \cdot A_{13} = 0$$

$$\Leftrightarrow x \begin{vmatrix} x & 2 \\ 2 & x \end{vmatrix} = 0$$

$$\Leftrightarrow x (x^2 - 4) = 0$$

$$\Leftrightarrow \begin{cases} x = 0 \\ x = \pm 2 \end{cases}$$

Câu 3:

$$\begin{pmatrix} x+y & 1 \\ -1 & x-y \end{pmatrix} \begin{pmatrix} 2 \\ 1 \end{pmatrix} = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$$

$$\Leftrightarrow \begin{pmatrix} 2(x+y) + 1 \\ -2 + (x-y) \end{pmatrix} = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$$

$$\Leftrightarrow \begin{cases} 2(x+y) + 1 = -1 \\ (x-y) - 2 = 1 \end{cases}$$

$$\Leftrightarrow \begin{cases} 2x + 2y = -2 \\ x - y = 3 \end{cases} \Leftrightarrow \begin{cases} x = 1 \\ y = -2 \end{cases}$$

Câu 5

$$A = \begin{pmatrix} 1 & 1 & 0 \\ 2 & -2 & 3 \\ -1 & 2 & 2 \end{pmatrix}$$

$$\begin{aligned} a, \det(A^5 - 2A^4) \\ &= \det A^4 (A - 2I) \\ &= \det A^4 \cdot \det(A - 2I) \end{aligned}$$

$$\text{Ta có } A^2 = A \cdot A = \begin{pmatrix} 1 & 1 & 0 \\ 2 & -2 & 3 \\ -1 & 2 & 2 \end{pmatrix} \cdot \begin{pmatrix} 1 & 1 & 0 \\ 2 & -2 & 3 \\ -1 & 2 & 2 \end{pmatrix}$$

$$= \begin{pmatrix} 3 & -1 & 3 \\ -5 & 12 & 0 \\ 1 & -1 & 10 \end{pmatrix}$$

$$\begin{aligned} A^4 &= A^2 \cdot A^2 = \begin{pmatrix} 3 & -1 & 0 \\ 5 & 12 & 3 \\ 1 & -1 & 10 \end{pmatrix} \cdot \begin{pmatrix} 3 & -1 & 3 \\ 5 & 12 & 0 \\ 1 & -1 & 10 \end{pmatrix} \\ &= \begin{pmatrix} 17 & -18 & 39 \\ -75 & 148 & -45 \\ 18 & -23 & 103 \end{pmatrix} \end{aligned}$$

$$A - 2I = \begin{pmatrix} 1 & 1 & 0 \\ 2 & -2 & 3 \\ -1 & 2 & 2 \end{pmatrix} - 2 \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} -1 & 1 & 0 \\ 2 & -4 & 3 \\ -1 & 2 & 0 \end{pmatrix}$$

$$\det A^4 = 83521$$

$$\det(A - 2I) = 3 \rightarrow \det A^4 \cdot \det(A - 2I) = 250563$$

b,

Ta có $\det A = -17 \neq 0 \rightarrow$ tồn tại A^{-1}

$$A_{11} = (-1)^2 \cdot \begin{vmatrix} -2 & 3 \\ 2 & 2 \end{vmatrix} = -10$$

$$A_{12} = (-1)^3 \cdot \begin{vmatrix} 2 & 3 \\ -1 & 2 \end{vmatrix} = -7$$

$$A_{13} = (-1)^4 \cdot \begin{vmatrix} 2 & -2 \\ -1 & 2 \end{vmatrix} = 2$$

$$A_{21} = (-1)^3 \cdot \begin{vmatrix} 1 & 0 \\ 2 & 2 \end{vmatrix} = -2$$

$$A_{22} = (-1)^4 \cdot \begin{vmatrix} 1 & 0 \\ -1 & 2 \end{vmatrix} = 2$$

$$A_{23} = (-1)^5 \cdot \begin{vmatrix} 1 & 1 \\ -1 & 2 \end{vmatrix} = -3$$

$$A_{31} = (-1)^4 \cdot \begin{vmatrix} 1 & 0 \\ -2 & 3 \end{vmatrix} = 3$$

$$A_{32} = (-1)^5 \cdot \begin{vmatrix} 1 & 0 \\ 2 & 3 \end{vmatrix} = -8$$

$$A_{33} = (-1)^6 \cdot \begin{vmatrix} 1 & 1 \\ 2 & -2 \end{vmatrix} = -4$$

$$\text{Ta có } A^{-1} = \frac{1}{\det A} \cdot A^* = \frac{1}{-17} \cdot \begin{pmatrix} -10 & -2 & 3 \\ -7 & 2 & -8 \\ 2 & -3 & -4 \end{pmatrix}$$

Câu 6:

$$\begin{pmatrix} 2 & 4 & -1 & 5 \\ -4 & -5 & 3 & -8 \\ 2 & -5 & -4 & 1 \end{pmatrix} \quad \begin{array}{l} H_1 (+2) + H_2 \rightarrow H_2 \\ H_1 (-1) + H_3 \rightarrow H_3 \end{array}$$

$$\rightarrow \begin{pmatrix} 2 & 4 & -1 & 5 \\ 0 & 3 & 1 & 2 \\ 0 & -9 & -3 & -4 \end{pmatrix}$$

$$\rightarrow \begin{pmatrix} 2 & 4 & -1 & 5 \\ 0 & 3 & 1 & 2 \\ 0 & 0 & 0 & 2 \end{pmatrix}$$

$$\Rightarrow \lambda(A) = 3$$

Câu 4,

$$VD: A = \begin{pmatrix} 1 & -1 \\ 1 & -1 \end{pmatrix}$$

$$A^2 - A \cdot A^T = \begin{pmatrix} 1 & -1 \\ 1 & -1 \end{pmatrix} \cdot \begin{pmatrix} 1 & -1 \\ 1 & -1 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$$